

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

CS2A - Risk Modelling and Survival Analysis

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) The following time series model is used for modelling medical inflation (X_t),

$$X_t = 0.6X_{t-1} + 0.3X_{t-2} + Z_t + 0.04,$$

where $\{Z_t\}$ is a sequence of uncorrelated identically distributed random variables, whose distributions are normal with mean zero. Using ARIMA (p, d, q) model produce the values of p , d and q .

A. (1,0,0)

B. (0,1,0)

C. (2,0,0)

D. (0,2,0)

E. (0,0,1)

[2 Marks]

Q.2) Write down the below matrix form into AR model form

$$\begin{bmatrix} X_n \\ X_{n-1} \\ X_{n-2} \end{bmatrix} = \begin{bmatrix} 3 & 3 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_{n-1} \\ X_{n-2} \\ X_{n-3} \end{bmatrix} + \begin{bmatrix} e_n \\ 0 \\ 0 \end{bmatrix}$$

A. $X_n = 3X_{n-1} + 3X_{n-2} - X_{n-3} + e_n$

B. $X_n = X_{n-1} + e_n$

C. $X_n = X_{n-2} + e_n$

D. $X_n = 3X_{n-1} + X_{n-3} + e_n$

E. $X_n = 3X_{n-1} + X_{n-2} + e_n$

[2 Marks]

Q.3) What is the value of autocorrelation co-efficient with lag 1 for the timeseries process, $X_t = 0.08X_{t-1} + 0.49X_{t-2} + 0.39X_{t-3} + \varepsilon_t$, where ε_t is white noise

A. 0.9235

B. 0.8781

C. 0.8298

D. 0.9086

E. 0.1620

[2 Marks]

Q.4) A Markov jump process has two states, with force of transition from state 1 to 2 is 0.03 and from state 2 to 1 is 0.05. The value of Kolmogorov's forward differential equation for $p_{12}(t)$ is

- A. $\frac{2-e^{-0.08t}}{8}$
- B. $\frac{3}{8}(1 - e^{-0.08t})$
- C. $\frac{2}{8}(1 - e^{-0.08t})$
- D. $\frac{3}{8}(1 + e^{-0.08t})$
- E. $\frac{3-e^{-0.08t}}{8}$

[2 Marks]

Q.5) A Customer care unit of a large Bank, receives call from its customers, following poisson process with a mean rate of two calls per second. Calculate the probability of receiving at least 120 calls in first one minute, given that it has already received 115 calls in the first 55 seconds?

- A. 0.9707
- B. 0.0527
- C. 0.9897
- D. 0.0166
- E. 0.9329

[2 Marks]

Q.6) Evaluate if the below process is both stationary and invertible?

$$8X_n = 10(X_{n-1} - e_{n-1}) - 3(X_{n-2} - e_{n-2}) + 8e_n$$

- A. Stationary & non-invertible
- B. Non-stationary & non-invertible
- C. Non-stationary & invertible
- D. Stationary & invertible
- E. None of the above

[2 Marks]

Q.7) Types of variables which could be modelled using Fundamental Copulas are,

- A. To be independence or perfect positive independence
- B. To be independence or perfect negative independence
- C. Perfect positive independence and perfect negative independence only
- D. All of the above
- E. None of the above

[2 Marks]

Q.8) If the expected value of X_t is 2.25 for the stationary timeseries equation, $X_t = 0.9 + nX_{t-1} + 0.2X_{t-2} + e_t$. What is the value of “n”?

- A. 1.2
- B. 1.6
- C. 0.6
- D. 0.4
- E. 0.8

[2 Marks]

Q.9) Using transition probabilities, a Health insurer is trying to model the state of patients (Healthy, hospitalized in normal ward, hospitalized in ICU). The Chapman-Kolmogorov used should ensure

- A. Uniform transition probabilities
- B. Constant transition matrices
- C. Probabilities are consistent across multiple time steps
- D. Recovery time to be deterministic
- E. The historical transition probabilities

[2 Marks]

Q.10) An insurer is trying to graduate the mortality rates using mortality function

$$\mu_x = 0.001 * (1 + \frac{1.2^x}{10})$$

What would be the chi-square test statistic value for the sample data given below?

Sample Ages	Exposed to risk	Deaths reported
30	15932	395
31	15836	460
32	15740	555

- A. 0.096723
- B. 0.102097
- C. 0.107470
- D. 0.112844
- E. 0.118217

[2 Marks]

Q.11) If the mortality rates for the sample ages given below assume to follow Makeham's Law, what would be the estimated mortality rate for age 60?

$$\mu_{30} = 0.034563$$

$$\mu_{35} = 0.045167$$

$$\mu_{40} = 0.059224$$

- A. 0.178722
- B. 0.064907
- C. 0.160849
- D. 0.058416
- E. 0.151914

[2 Marks]

Q.12) On a short-term accident policy, the claim experience is expected to follow Pareto distribution with parameters $\alpha = 1.86625$ and λ . In a sample of 50 claims reported last year, it had a lower quartile of 25. What would be the value of λ ?

- A. 100

- B. 125
- C. 135
- D. 150
- E. 175

[2 Marks]

Q.13) The survival function $S_x(t)$ for life aged x , is calculated by,

- A. $S_x(t) = \int_0^t s \cdot \mu_{x+s} ds$
- B. $\ln S_x(t) = - \int_0^t \mu_{x+s} ds$
- C. $\ln S_x(t) = \int_0^t \mu_{x+s} ds$
- D. $\ln S_x(t) = - \int_0^t s \cdot \mu_{x+s} ds$
- E. $\ln S_x(t) = - \int_x^t \mu_{x+s} ds$

[2 Marks]

Q.14) There are two random variables (X, Y) represents lifetime of two batteries used in a toy car which is used for 30 minutes already.

$$P(X \leq 30) = 0.37845 \text{ and } P(Y \leq 30) = 0.30175.$$

Calculate the value of joint probability of both batteries don't last for another 30 more mins, using Frank Copula with $\alpha=3$.

- A. 0.017774
- B. 0.108570
- C. 0.184681
- D. 0.461562
- E. 0.118150

[2 Marks]

Q.15) On an existing protection product portfolio of an insurer, the number of claims expected to follow Poisson distribution with mean 12 and claim amount to follow pareto distribution ($\alpha = 2.25$) with mean 30. What would be the value of variance of the aggregate claim he might settle?

- A. 1,08,000
- B. 97,560
- C. 35,100
- D. 38,400
- E. 38,333

[2 Marks]

Q.16)

- i) Define the difference between strictly stationary processes and weakly stationary processes. [1]
- ii) When does the strictly stationary process and weakly stationary process are equivalent. State with reasoning? [1]
- iii) A study of timeliness of arrival of Metro train at a particular station was performed using 250 samples and given below are the SACF (sample autocorrelation function) values:
- $$r_1 = 0.042, r_2 = 0.191, r_3 = 0.023, r_4 = -0.085, r_5 = -0.021, r_6 = 0.032$$
- Test at 5% level, if the above data considers to be a white noise using Portmanteau test. [3]
- iv) Derive the Yule-Walker equations for the autocovariance function (ACF) at lags 0, 1, 2 and >2 for the models given by, $X_t = a_1X_{t-1} + a_2X_{t-2} + b_1e_{t-1} + e_t$.
- where e_t is a white noise process with variance σ^2
- Hint:** [For a white noise process, covariance (e_t, e_{t-s}) = 0 for $s \neq 0$.] [4]
- v) For a stationary AR (2) process defined by the equation below, derive the value of ρ_k for lag $k = 0, 1, 2$ and >2 ?
- $$X_t = \frac{9}{20}X_{t-1} - \frac{1}{20}X_{t-2} + e_t$$
- [5]
- [14 Marks]

Q.17) In Indiana, every year, “Indiana Premier Chess Competition (IPCC)” is conducted by Board of Chess Federation of Indiana, where all the chess professionals take part in the premier event. Due to its popularity, all the Sports Clubs across the country, provide sponsorship for those who like to represent the competition on behalf of their Club, to promote the Chess Competition.

A leading Sports Club “Sports & Games” runs from Sothern part of Indiana, announced a sponsorship programme for funding one chess player to take part in the IPCC event, invites

players to take part in their selection process, three months prior to the IPCC event. It conducts selection process at two levels,

- Conducts initial selection rounds to select top three players
- In the final selection round, those three players qualified from Initial round, would play among them to select one player, who will represent the Club in the competition

Note: In the world of competitive chess, a "rapid fire" game usually refers to Rapid Chess, a format where the time control is faster than standard "classical" chess but slower than the frantic pace of chess games.

Mr. Adithya Anand is one of the top three chess players, who has participated in the final selection round and below is his standing on the daily sprints conducted over 24 days. Ranking of "1" denote lowest standing and "3" being the highest standing, among the three players.

Day	1	2	3	4	5	6	7	8	9	10	11	12
Ranking	1	2	3	1	3	3	2	1	1	2	1	1
Day	13	14	15	16	17	18	19	20	21	22	23	24
Ranking	2	1	3	2	1	1	3	2	2	2	1	2

He is one of your best friends, understands that you are good in developing statistical models, approached you to evaluate his likelihood of being selected to represent the Club in IPCC event. You plan to use Markov chain 3-state model which could help assess his chance of he getting selected by the Club.

i) Calculate the values of,

a) n_{ij} [2]

b) n_i [1]

ii) Estimate the one-step transition probabilities of 3X3 matrix. [2]
[5 Marks]

Q.18) A drug company called "IMORTAL UNLIMITED", engaged in research activities, aiming to produce drugs for all the major life-threatening disease at different ages. The research team has observed that the major cause of life-threatening diseases varies by age and hence it found it to be more complex to produce one drug that could improve the longevity of human lives. Hence, they decided to tie-up with a large Health Insurance company called "LIVE FOREVER" to tap into their database and analytics expertise to get a direction to work on this matter.

The team has prioritized focusing on the age 30 demographic for maximum commercial viability. Analytics show this cohort is uniquely defined by mental health risks and early metabolic (BP/Sugar) shifts rather than chronic disease. This age represents a critical "biological trigger" point; chronic inflammation from high stress and sedentary lifestyles at 30 often sets the foundation for major medical events in the future ages. According to the analysis, here are the estimated transition rates between the states for a male aged 30, which is used in evaluating the effectiveness of the drugs produced for a given risk:

The defined states are, H – Healthy; Mi – Mental issues; Me – Metabolic issues; Ar – Accident / rest issues; D - Death

States of transition	Transition rate
Healthy (H) to Death (D) directly	0.019
Healthy to Mental Health issues (Mi)	0.032
Healthy to Metabolic diseases (Me)	0.021
Healthy to Accidents/rest causes (Ar)	0.016
Mental Health issues to Death	0.6
Metabolic diseases to Death	0.8
Accident/rest major causes to Death	0.4

- i) Prepare a transition matrix that captures movement between the states during the year with annual constant transition rate, that represents a Markov jump model of the process, using the above information given. [2]
- ii) Calculate the probability of a healthy male life aged exactly at 30 dies by the end of one year time, using above transition rates through each state? [9]

Given Accidents/Trauma and Cancers are less of worry for this age group, the team has decided to focus on metabolic issues and Mental Health issues only, to improve the long-term wellness of this age group. After intensive research and taking necessary approvals, the Drug company has decided to introduce a new drug in the market which it claims that it could help reduce the transition rate by 40% from Mental health issues and 60% of those suffering from Metabolic diseases leads to death. The change in probabilities are as mentioned below:

Transitions	Revised probability
Healthy to dead which had Mental issues	0.004975
Healthy to death which had Metabolic issues	0.002939

- iii) Assess impact of introduction of the new drug on each of the transitions from each state and overall probability of death too and comment on the results. [3]

[14 Marks]

Q.19) A Health Insurer is pricing a new hospital cash product and it's trying come up with the assumption on length of stay required for an insured admitted into the hospital for a particular illness, to incorporate the same in the pricing basis. Referring to investigations done by various medical research firms, it understands that the level of **Lifestyle disorders**, leads to different level of impact on the duration of recovery of any specific health conditions.

Lifestyle disorders—often called "diseases of civilization"—stem from sedentary habits, poor diet, and unhealthy routines. According to research done by the firm, Lifestyle disorders impact human system leading to various disorders/diseases in our body, namely,

- **Metabolic & Endocrine issues** - Type 2 Diabetes, Obesity, Metabolic Syndrome
- **Cardiovascular issues** – Hypertension, Dyslipidemia, Atherosclerosis
- **Respiratory & Bone Health issues** - Chronic Obstructive Pulmonary Disease, Osteoporosis

The researchers of the view that the key influencing factors of the above are, **Diet, physical activity, sleep** and consumption of **Tobacco & alcohol**.

As it is quite complex to understand the interplay of all these factors, the insurer has decided to use Cox proportional hazard model to analyze the recoverability of an individual from lifestyle disorders and it has come up with the parameters as mentioned below:

Covariate	Category	Parameter
Smoking	Non-smoker	-0.025
	Smoker	0.104
Diet & Exercise	Regular	0
	Irregular	0.391
	Regularly eats from outside, no exercise	0.459
Alcohol & Sleep	non-alcoholic, sleeps 8 hours	-0.046
	Drinks once a month, sleeps 6 hours	0
	Heavy drinking, Night shift	0.378

- i) Develop the hazard function using above parameters in the Cox proportional hazards model, stating all necessary details of the components used in the function? [4]

From the researches it is observed that, patients aged below 30, who falls under “non-smoker, regular on diet & exercise, non-alcoholic, sleeps for 8 hours” category, have 80% probability of recovering from illness within two days of hospital stay.

- ii) Using the above information, can you calculate the probability of length of hospital stay required for more than two days, for the following cases? [4]

- a) Smoker - regularly eats from outside, no exercise - Heavy drinking, Night shift
- b) Non-smoker, irregular on diet & exercise, but non-alcoholic, sleeps 8 hours

Upon reviewing the calculations, the pricing actuary recommended a few changes in the model, to split both Diet and Exercise separately and add age also as one of the covariates to derive the probabilities appropriately.

Covariate	Category	Parameter
Smoking	Non-smoker	-0.025
	Smoker	0.104
Diet	Regular	0

	Irregular	0.198
	Regularly eats from outside	0.234
Exercise	Regular	0
	Irregular	0.193
	No exercise	0.225
Alcohol & Sleep	non-alcoholic, sleeps 8 hours	-0.046
	Drinks once a month, sleeps 6 hours	0
	Heavy drinking, Night shift	0.378
Age	below 30	-0.053
	30-45	0
	45 and above	0.032

iii) It was expected that the probability of recovering from illness within two days of hospital stay, used in part (ii) increases by 10% for younger age group. Re-calculate the probabilities given in (ii) using the above additional information for the “age band below 30”? [4]

iv) Compare and comment on the two results obtained from part (ii) and (iii). [2]

[14 Marks]

Q.20) The Chief Production Officer of a large cement factory observes that the productivity of the cement factory has considerably reduced over the last three months. Hence, he has asked his team to investigate all the processes in detail to figure out the cause of the reduction. The issues were found with the failure of feeder ducts at multiple sections which cause wastage of materials during the process. The life time of feeder ducts in a cement factory have guarantee period of 2 years for repair and replacement. The company observes replacement of the following feeder ducts of the total 20 feeder ducts.

Month	Failure of feeder ducts
6	1 duct in "Kiln 1"
11	2 ducts in "Kiln 2"
14	1 duct in "Silo 1" due to major accident in the factory
15	2 ducts in "Silo 2"
17	1 duct in Limestone work yard
20	1 duct in "Kiln 3" due to wrong handling
22	1 duct in the package yard

The team has reported all types of failures - intrinsic as well as others as can be seen in the table. However, while calculating the estimates of the survival function, only intrinsic failures are to be considered and failures due to external factors are to be ignored.

- i) Use Kaplan-Meier method to estimate the distribution function of the lifetime for the feeder ducts. [5]
- ii) Use Nelson-Aalen to calculate the cumulative hazard function of the feeder ducts and hence estimate the survival function of the feeder ducts. [3]

[8 Marks]

Q.21) A small life insurer who primarily writes pure protection business, offers cover on fixed sum assured basis to a selected target population. These policies are typically issued for longer periods (typically, 20-40 years), where the insurer is obliged to cover the risk as long as the policyholders continue to pay premiums as per product design during the coverage period.

The first version of the product (Product A) was launched at the beginning of the year 2016 by the insurer. Below are the details about the in-force policies (i.e., the policy is in active state and insurer covers the risk) including other details such as business mixes and claim experience:

Fixed Sum assured of Product A	16	32	64
Proportion of business mix	40%	35%	25%
Expected Claim experience	<i>Pareto(6, 0.5)</i>	<i>Pareto(5.5, 0.75)</i>	<i>Pareto(5, 1)</i>

- The annual lapse rate of the product is 5%, and for the purpose of the problem, assume there are no other decrements.
 - Assume the sum at risk under each policy is equal to sum assured for the calculation purposes.
 - At the end of year 2020, insurer stopped issuing new policies under product A, and 100 policies were in-force at that time and continue to cover the existing in-force policies.
 - Claim experience assumed to follow *Pareto(α, λ)* distribution with parameters as mentioned above.
- i) The Insurer had executed 50:50 quota-share reinsurance arrangement and it is in active state. Calculate the expected amount of total claim likely to be settled by the re-insurer considering in-force risk at present, under the Quota-share arrangement in the end of year 2025. [3]
 - ii) If the insurer were to move from Quota-share to surplus reinsurance arrangement, what would be the value of retention limit which will lead to total amount of risk reinsured equal between the two methods. [3]

Insurer has undertaken a review of claim experience for product A, benefit assessment of reinsurance arrangement and solvency position. Based on review, the Insurer has updated its

underwriting norms, repriced and launched new version (Product B) of the product in the year starting 2021, to replace Product A.

Insurer has taken conscious decisions initially to retain the risks underwritten internally over the years for various financial reasons. But given the risk written was more than anticipated, the insurer now has decided to execute a surplus arrangement for Product B to handle claim volatility effectively. It aims to set retention limit for Product B, such that it retains 60% of total in-force risk from both products combined together.

Use the following information to determine the retention limit for Product B:

- Ultimate level of claim experience to be assumed for the purpose of reinsurance calculations
- New business written under Product B from the launch, is as given below by each year;

2021	2022	2023	2024	2025
25	30	40	55	60

- The annual lapse rate of the product is 10%, and for the purpose of the problem, assume there are no other decrements.
- Assume sum at risk under each policy is equal to sum assured.
- The proportion of average sum assured written as follows:

Fixed Sum assured of Product B	16	32	64
Proportion of business mix	30%	35%	35%

- iii) Calculate the retention limit to be used for reinsuring policies under Product B at the end of year 2025, considering the existing reinsurance arrangement for Product A? [5]

The insurer has now decided to terminate the reinsurance arrangement executed for product A as it is closed to new business and existing business is running off gradually. This is leading to a higher retention of risk, and thus has decided to use retention limit of 10 on Product B to finalize the reinsurance arrangement.

- iv) For the surplus reinsurance arrangement, the Reinsurer has used log-normal distribution with $\mu=2.6$ and $\sigma= 0.8$ to price the reinsured risk. Calculate the expected claim size which the reinsurer likely to settle for the revised retention limit. [4]

[Hint:

1. Use standard normal variable,

$$Z = \frac{(\ln(x) - \mu)}{\sigma}$$

to transform from x,

2. Substitute z in below equation and convert it into standard normal distribution

$$\int_M^{\infty} x \cdot f(x) dx$$

3. Use conditional probability to derive the solution]

[15 Marks]
